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A causal variational worldline action for modified inertia: the rapidity gap

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Version 3 (2026-08-07). Changes from v1 and v2 are listed in §10.

Abstract

Modified-inertia (MI) realisations of the MOND phenomenology require an inertial mass that depends on the magnitude of the proper acceleration, $|a|$ — a non-analytic function of the acceleration squared. No polynomial-in-velocity worldline self-interaction can produce it. We prove this as a parity theorem: because Minkowski space admits only the rank-2 metric and the rank-4 Levi-Civita tensor as invariants, every polynomial scalar built from four-velocities at several proper times has even degree in u and therefore contributes only even powers of the orbital speed, whereas the deep MOND limit requires the first power. We then exhibit the unique escape class. The square root of the velocity bilinear is exactly the **rapidity gap** between the velocities at two proper times, $\sqrt{(-u \cdot u/c^2 - 1)} = \sqrt{2} \sinh(\theta/2)$ with $\cosh \theta = -u \cdot u/c^2$, and it is linear in the speed; its short-separation limit is $\theta(s) \rightarrow |a|s/c$, because proper acceleration is the rate at which rapidity accumulates. A retarded kernel smearing of θ therefore delivers $|a|$ with no (v/c) suppression.

With the minimal causal kernel $K(s) = (N/\lambda)e^{-(s/\lambda)}$ the model reduces, on circular worldlines, to the closed form $\Theta = (4Nv/c) \cdot x \cdot \coth(\pi/x)/(4+x^2)$ with $x = \lambda\Omega$, and the circular-orbit equation of motion is $\bar{g} = \mu(g_{\text{obs}}/a_0) \cdot g_{\text{obs}}$ — Milgrom's modified-inertia relation exactly, with $\mu = 1 + F$. Demanding the interpolation exponent that solar-system ephemerides require ($\alpha \geq 1.4$; $\alpha = 1$ is excluded) gives F in closed form. The construction is causal, variational in the in-in sense, free of Ostrogradsky instability, and Lorentz-invariant up to the preferred frame that MI already presupposes.

We identify and repair its worst defect. In a single-factor worldline action the same function

multiplies the rest energy and the inertia, so $\mu \rightarrow 0$ in the deep limit forces the rest energy to vanish. We prove this is structural — every $d\tau W(\Theta)$ term locks the two at a ratio independent of W — and repair it with a two-function form built on the preferred-frame four-velocity n that MI already presupposes. The correct covariant object is $\sqrt{((u \cdot n)^2)}$, and with it the repaired action has rest energy exactly mc^2 , inertia exactly $m\mu$, energy $E = mc^2[1 + \mu(\gamma-1)]$ bounded below at every speed, and CPT-even structure. A linear-in- $(u \cdot n)$ writing agrees with it on every particle worldline but continues differently to antiparticles, where it gives rest energy $mc^2(2\mu-1)$ — negative for $g < 0.52 a_0$, i.e. throughout the outer-galaxy regime — and is therefore excluded. The surviving preferred-frame coupling scales as g^{-2} , so the Lorentz violation is largest in the lowest-acceleration environments and $\sim 10^{-23}$ in a terrestrial laboratory.

General orbits are then solved. For an arbitrary worldline $\theta(\tau, \tau-s) = (s/c)|a(\tau - s/2)| + O(s^3)$ — the rapidity gap across an interval is s/c times the acceleration magnitude at the *midpoint* — so Θ is a retarded memory-average of $|a|$ with lag $s/2$. The equation of motion acquires a **memory force**, $d/dt[m\mu v] = -m\nabla\Phi + (m/2a_0)d^2/dt^2[\mu v^2\hat{a}]$, which does **not** vanish on circular orbits: the correct balance is $g_bar = g_obs[\mu + (Y/2)\mu]$. MOND survives this as a pure renormalisation $a_0 \rightarrow (2/3)a_0$, and any target interpolation remains exactly reachable. The memory force also cancels the leading Newtonian anomaly identically at $\alpha = 2$, suppressing the solar-system residual by a further factor $8Y^2 = 3 \times 10^{16}$.

We are explicit about what is not achieved: the acceleration scale is not derived. The kernel's first moment must satisfy $M_1 = c/a_0$, and in the short-memory limit only that moment survives — verified across three kernel shapes — so one number is traded for one number. **The coefficient of a_0 remains fitted, not derived.**

1. Introduction and prior art

MOND phenomenology is organised by a single acceleration scale $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$ and an interpolating function between the Newtonian and deep regimes. Two realisations are distinguished: modified gravity (MG), in which the field equations change, and modified inertia (MI), in which the particle's inertial response changes. This paper is about MI.

Credit, stated up front and without hedging. The interpolating kernel used throughout, $\nu(y) = \sqrt{1 + 1/y}$ with $y = g_bar/a_0$, is *identically* Milgrom's kernel — M. Milgrom, *Phys. Lett. A* **253** (1999) 273, equations 6–9 — obtained there from a de Sitter–Unruh temperature balance $T = \sqrt{(a^2 + \Lambda c^2/3)}/2\pi$, with the coefficient fixed to $\hat{a}_0 = 2cH_ \Lambda$. His equations 10–11 give a second coefficient, and Milgrom 2008 §7.3.1 records that the mismatch between them “isn't necessarily meaningful.” The scale $a_ \lambda = c^2 \sqrt{(\Lambda/3)}$ is Milgrom, *Ann. Phys.* **229** (1994) 384. The temperature $\sqrt{(a^2 + \Lambda/3)}/2\pi$ is Narnhofer, Peter and Thirring, *IJMPB* **10** (1996) 1507. Nothing in the kernel is new here.

What is new here is the *action*: a causal, variational worldline functional whose equation of motion is the MI relation, together with the theorem that identifies which structure can produce it.

What is not claimed. This paper does not derive a_0 , does not derive its coefficient, and makes no claim about unification or the Standard Model. Section 8 states precisely which single number is fitted and why the construction cannot fix it.

2. The obstruction: a parity theorem

Consider a worldline $x^\mu(\tau)$ with $u^\mu = dx^\mu/d\tau$, $u \cdot u = -c^2$. A nonlocal self-interaction built from velocities at several proper times is a candidate for an MI action. Take first the bilinear form.

On an exact circular worldline of speed v and coordinate angular velocity Ω , writing $\gamma = (1-v^2/c^2)^{-1/2}$,

$$(1) \quad u(\tau) \cdot u(\tau+s) = -\gamma^2 c^2 [1 - (v^2/c^2) \cos(\gamma \Omega s)]$$

which is independent of τ and depends on the trajectory only through v and $\gamma \Omega$. Consequently, smearing with any kernel $K(s)$ returns a constant plus $(v^2/c^2) \times$ (a function of $\gamma \Omega$). But the quantity MOND needs is the proper acceleration magnitude,

$$(2) \quad |a| = \gamma^2 \Omega v,$$

which is **first** order in v . The mismatch is exactly one power of v ; in amplitude it is $(c/v)^2$, which for galactic speeds of 50–300 km s⁻¹ is 10^6 – 3.6×10^7 . This reproduces, from power counting alone, the suppression factor found by direct computation in the modified-inertia programme.

Higher polynomial order does not help, and this is a theorem rather than a survey:

Theorem 1 (parity). Minkowski space admits exactly two invariant tensors, the metric η (rank 2) and the Levi-Civita tensor ε (rank 4), both of **even** rank. Every polynomial scalar constructed from four-velocities at any number of proper times therefore has even total degree in u , and on a circular worldline contributes only **even** powers of v . No polynomial-in- u worldline self-interaction, at any degree, can produce the deep-MOND v^1 scaling.

There is no rank-1 or rank-3 invariant, so an odd-degree scalar self-interaction does not exist to be attempted. Direct check: the quartic $(-u \cdot u/c^2 - 1)^2$ has short-separation limit $|a|^4 s^4 / 4c^4$ — degree four in v , moving further from the target.

3. The escape: the rapidity gap

Theorem 1 bans polynomials. It does not ban non-analytic functions of the bilinear, and there is a canonical one. From (1),

$$(3) \quad -u \cdot u/c^2 - 1 = 2\gamma^2 (v^2/c^2) \sin^2(\gamma \Omega s/2)$$

so that

$$(4) \quad \sqrt{-u \cdot u/c^2 - 1} = \gamma(v/c) \cdot |\sin(\gamma \Omega s/2)|$$

which is **first** degree in v . Geometrically, since $\cosh \theta = -u \cdot u/c^2$ defines the hyperbolic angle (rapidity gap) between two unit timelike vectors, and $\cosh \theta - 1 = 2 \sinh^2(\theta/2)$,

$$(5) \quad \sqrt{-u \cdot u/c^2 - 1} = \sqrt{2} \sinh(\theta/2) \text{ exactly.}$$

θ is a Lorentz scalar. Its short-separation behaviour is the point of the construction:

$$(6) \theta(s)/s \rightarrow |a|/c, \sqrt{(-u \cdot u/c^2 - 1)/s} \rightarrow |a|/(\sqrt{2} c) \text{ as } s \rightarrow 0.$$

The first of these is the classical statement that **proper acceleration is the rate at which rapidity accumulates along a worldline**. So $|a|$ appears linearly, with no (v/c) suppression. The proposed action is a functional of the accumulated rapidity:

$$(7) S = -mc^2 \int d\tau \sqrt{(1 - v^2/c^2)} [1 + F(\Theta)], \Theta(\tau) = \int_0^\infty ds K(s) \theta(\tau, \tau-s)$$

with the kernel supported on $s > 0$ only, so the construction is **causal** by fiat. Only $u = dx/d\tau$ appears, at several proper times: there is no second or third derivative of x anywhere, hence **no Ostrogradsky instability**. With a retarded kernel this is precisely the nonlocal worldline class for which the in-in (Schwinger–Keldysh) functional is variational, so (7) is variational in that sense.

4. A kernel, and the closed form

$\theta(s)$ is **periodic** in s with period $2\pi/(\gamma\Omega)$ and bounded above by twice the rapidity, $2 \operatorname{arctanh}(v/c)$. It behaves as $|a|s$ only for $s \ll 1/\Omega$. Which regime obtains is therefore the whole content of the model.

Take the minimal causal one-scale kernel $K(s) = (N/\lambda) e^{-(s/\lambda)}$. Using $\int_0^\infty e^{-(ps)} |\sin ks| ds = k(p^2 + k^2)^{-1} \coth(\pi p/2k)$, the non-relativistic reduction is exact:

$$(8) \Theta(v, \Omega) = (4N v/c) \cdot x \coth(\pi/x) / (4 + x^2), x \lambda \Omega$$

(verified against numerical quadrature to 20 digits at $x = 0.05, 1, 20$). Its two limits are qualitatively different:

- **Short memory, $x \ll 1$:** $\Theta \rightarrow M_1 |a|/c$ with $M_1 \int K(s) s ds = N\lambda$. Setting $M_1 = c/a_0$ makes $\Theta = |a|/a_0$ *exactly* — the MOND variable itself, with no residual factor.
- **Long memory, $x \gg 1$:** $\Theta \rightarrow (4N/\pi)(v/c)$, a function of **speed**, with Ω absent.

5. The equation of motion

Non-relativistically (7) expands to $L \simeq -mc^2[1+F] + \frac{1}{2}m[1+F]v^2 - m\Phi$, so the inertial mass is $m_{\text{eff}} = m[1 + F(\Theta)]$ with $\mu = 1 + F$ and $\nu = 1/\mu$. Because Θ depends on the acceleration, the variation produces a second term. At leading order in $\lambda\Omega$ the variation of Θ is *local*, $\delta\Theta = (M_1/c) \hat{a} \cdot \delta a$ with $\hat{a} = a/|a|$, and one integration by parts on the first term and two on the second give

$$(9) d/dt [m \mu(\Theta) v] = -m\nabla\Phi + (m/2a_0) d^2/dt^2 [\mu(\Theta) v^2 \hat{a}]$$

The second term is the **memory force**. It is fourth order in x — generic for a nonlocal action, and *not* an Ostrogradsky instability, since that theorem requires a local higher-derivative Lagrangian (§9 item 3).

General orbits. For an arbitrary worldline, $u \cdot u = -c^2$ forces $u \cdot a = 0$, $u \cdot \dot{a} = -|a|^2$ and $u \cdot \ddot{a} = -3a \cdot \dot{a}$, whence $\cosh \theta - 1 = s^2 |a|^2/2c^2 - s^3 (a \cdot \dot{a})/2c^2 + O(s^4)$ and therefore

$$(10) \theta(\tau, \tau-s) = (s/c) |a(\tau - s/2)| + O(s^3)$$

The rapidity gap across an interval s is s/c times the acceleration magnitude at the **midpoint**, so

$$(11) \Theta(\tau) = (1/c) \int_0^\infty ds K(s) s |a(\tau - s/2)|$$

is a retarded memory-average of $|a|$ with weight $K(s)$ and lag $s/2$, collapsing to $|a|/a_0$ when $|a|$ varies slowly over λ . This is where the in-in prescription earns its place: varying a kernel that joins τ to $\tau-s$ places a factor evaluated at $\tau + s/2$ — the *future* — in the coefficient of $\delta u(\tau)$, so a retarded kernel in the action would otherwise give a time-symmetric equation of motion. The advanced terms enter only at $O(\lambda\Omega)$, which is $\leq 4 \times 10^{-6}$ galactically at $\lambda \leq 39$ yr (§8), so (9) is causal at the accuracy that matters and the in-in machinery is needed only for those corrections.

The circular reduction, corrected. On a circular orbit $\hat{a} = -r$ rotates, so $d^2\hat{a}/dt^2 = \Omega^2 r \neq 0$ and the memory force is outward with magnitude $(m/2a_0)\mu g_{\text{obs}}^2$. The balance is therefore

$$(12) g_{\text{bar}} = g_{\text{obs}} [\mu(Y) + (Y/2) \mu(Y)] g_{\text{obs}} \mu_{\text{eff}}(Y), Y = g_{\text{obs}}/a_0$$

not $g_{\text{bar}} = \mu g_{\text{obs}}$. In the deep limit $\mu \rightarrow Y$ gives $\mu_{\text{eff}} \rightarrow (3/2)Y$, i.e. $g_{\text{obs}}^2 = (2/3)a_0 g_{\text{bar}}$: **MOND survives as a pure renormalisation** $a_0 \rightarrow (2/3)a_0$, so the required kernel moment becomes $M_1 = (2/3)c/a_0$. The Newtonian limit $\mu_{\text{eff}} \rightarrow 1$ is preserved. And any target interpolation μ_t remains reachable exactly, since $\mu + (Y/2)\mu = \mu_t$ is a first-order linear ODE with solution

$$(13) \mu(Y) = (2/Y^2) \int_0^Y Y \mu_t(Y) dY$$

The long-memory branch is excluded **structurally**, not by data. A kinetic function of speed alone, $L = m f(v) - m\Phi$, gives circular-orbit balance $g_{\text{bar}} = g_{\text{obs}} f(v)/v$; matching the deep limit $g_{\text{obs}} = \sqrt{g_{\text{bar}} a_0}$ then requires $f(v) = v^3/(r a_0)$, which depends on r at fixed v . No such f exists. The model must therefore live in the short-memory regime, and this is a constraint on λ , not a choice.

Solving for F. Solar-system ephemerides bound a constant anomalous radial acceleration. In the α -family $\nu_\alpha(y) = (1 + y^{(-\alpha)})(1/2\alpha)$, equivalently $g_{\text{obs}} \hat{(2\alpha)} = g_{\text{bar}} \hat{(2\alpha)} + a_0 \hat{\alpha} g_{\text{bar}} \hat{\alpha}$, the Newtonian-limit anomaly is $\Delta = a_0 \hat{\alpha} g_{\text{bar}} \hat{(1-\alpha)}/(2\alpha)$. At $\alpha = 1$ this is the *constant* $\Delta = a_0/2$, which exceeds the bound by $\sim 10^3$; the bound requires $\alpha \geq 1.38$ before, and ≥ 1.25 after, the external-field effect. Taking $\alpha = 2$ and inverting $g_{\text{obs}}^4 = g_{\text{bar}}^4 + a_0^2 g_{\text{bar}}^2$, with $Y = g_{\text{obs}}/a_0$:

$$(14) \mu_2(Y) = \sqrt{(-1 + \sqrt{1 + 4Y^4}) / 2} / Y, F(Y) = \mu_2(Y) - 1$$

This is monotone, tends to 1 as $Y \rightarrow \infty$ (Newtonian recovery, $F \rightarrow 0$) and to Y as $Y \rightarrow 0$ (deep MOND, $\mu_{\text{eff}} \rightarrow m|a|/a_0$). The reproduction of ν_2 is exact by construction, so (14) is a **consistency** result — a solution F exists — not a prediction of the shape.

And the memory force cancels the Newtonian anomaly at $\alpha = 2$. For the α -family $1 - \mu = Y \hat{(-\alpha)}/2\alpha$, the memory term of (12) contributes $+Y \hat{(-\alpha)}/4$, so

$$(15) 1 - \mu_{\text{eff}} = Y \hat{(-\alpha)} (2 - \alpha) / (4\alpha)$$

which **vanishes identically at $\alpha = 2$** . For the exact $\alpha = 2$ kernel (14) the residual is $1/(32Y^4)$ rather than $1/(4Y^2)$, so the solar-system anomaly is suppressed by a further $8Y^2 = 3.2 \times 10^{16}$ at Earth, falling to $1.2 \times 10^{-35} \text{ m s}^{-2}$ — twenty-one orders under the bound. Two facts cut the other way and belong here: the $\alpha = 1$ floor **survives** as $\Delta = a_0/4$ rather than $a_0/2$, so the exact a_0 -line is still excluded (by $640\times$ rather than $1279\times$) and α_{min} softens only from 1.380 to 1.322; and for $\alpha > 2$ the residual in (15) changes **sign**, giving net repulsion, which is a further structural constraint on the exponent.

6. The rest-energy defect, and its repair

Equation (10) has $F(0) = -1$. In (7) the same factor multiplies rest energy and inertia, so the rest energy $mc^2[1+F]$ **vanishes** at zero acceleration. This is not a bad choice of F :

Theorem 2. Any reparametrisation-invariant worldline term $d\tau W(\Theta)$ expands to $dt W(1 - v^2/2c^2)$, so its rest-energy and inertia contributions are locked at a ratio **independent of W** . No single function can separate them.

The repair requires a second structure that is not of the form $d\tau \times (\text{scalar})$. MI already supplies one: inertia in MI is defined relative to a preferred frame — the cosmological / de Sitter rest frame — with unit timelike four-velocity n , $n \cdot n = -c^2$, so $u \cdot n = -\gamma c^2$. Two independent structures follow.

Three candidate second structures exist; only one survives, and the selection is forced rather than a matter of taste.

Form I (quadratic in $u \cdot n$).

$$(16) S = -mc^2 d\tau A(\Theta) + (m/c^2) d\tau B(\Theta) (u \cdot n)^2$$

gives rest energy $mc^2(A - B)$ and inertial mass $m(A + B)$ — two independent combinations. Imposing unmodified rest energy and MI inertia gives $A = (1+\mu)/2$, $B = (\mu-1)/2$, so the rest energy is exactly mc^2 for every Θ and the inertia is exactly $m\mu(\Theta)$, with $B \rightarrow 0$ as $\mu \rightarrow 1$: the added structure switches itself off in the Newtonian limit. It is CPT-even, because $(u \cdot n)^2$ is even. Its cost is that the exact energy $E = mc^2[A\gamma + B(2v^2/c^2 - 1)\gamma^3]$ becomes negative above $v_{\text{crit}}^2 = 2c^2/(3 - \mu)$, i.e. $0.8165 c$ in the deep limit.

Form II (linear in $u \cdot n$).

$$(17) S = -mc^2 d\tau \mu(\Theta) - d\tau (u \cdot n)[\mu(\Theta) - 1]$$

also gives rest energy exactly mc^2 and inertia exactly $m\mu$, with exact energy

$$(18) E = mc^2[1 + \mu(\gamma - 1)]$$

monotone in γ and bounded below by mc^2 at every speed — no instability anywhere. But a term linear in u contracted with a background vector is an SME a^μ structure, hence CPT-odd, and the consequence is not merely aesthetic. On the antiparticle branch the second term reverses sign, giving rest energy

$$(19) \bar{E} = mc^2(2\mu - 1)$$

which is **negative** for $\mu < \frac{1}{2}$. For the $\alpha = 2$ kernel $\mu_2 = \frac{1}{2}$ at $Y = 2/\sqrt{15} = 0.5164$, i.e. $atg = 0.5164a_{\{0\}} = 4.83 \times 10^{-11} \text{ m s}^{-2}$. **So the CPT-odd writing predicts negative-rest-energy antimatter throughout the outer-galaxy regime the theory exists to describe.** It is excluded by a structural catastrophe, not by any numerical bound.

Form III (the even root) — and it supersedes both. For a future-directed timelike worldline $u \cdot n = -\gamma c^2 < 0$, so $(u \cdot n) = -|u \cdot n|$ identically, and (17) is *the same action as*

$$(20) S = -mc^2 d\tau \mu(\Theta) - m d\tau \sqrt{(u \cdot n)^2} [1 - \mu(\Theta)]$$

on every particle trajectory. The two writings differ only in the continuation to past-directed worldlines. Since $\sqrt{(u \cdot n)^2}$ is manifestly even under $u \rightarrow -u$, (20) is **CPT-even**, and therefore:

- rest energy exactly mc^2 and inertia exactly $m\mu(\Theta)$, as in both other forms;

- energy (18), monotone and bounded below by mc^2 at **every** speed — so Form I's 0.8165 c instability is not merely displaced but absent;
- antiparticles carry the same rest energy and the same inertia, so (19) does not arise;
- the preferred-frame structure is $c^{\wedge}\{\mu\nu\}$ -type (CPT-even) rather than $a^{\wedge}\mu$ -type.

The CPT-oddness of Form II was therefore an artefact of the writing, not a property of the theory, and Form I's instability was an unnecessary price. (20) is adopted for the remainder of this paper.

7. A prediction: g^{-2} preferred-frame coupling

For the $\alpha = 2$ kernel, $\mu_2 = 1 - a_0^2/4g^2 + O(g^{-4})$, so the strength of the preferred-frame term in (20) is

$$(21) \quad (1 - \mu)/2 \simeq a_0^2/(8g^2)$$

Evaluated with $a_0 = 9.36 \times 10^{-11} \text{ m s}^{-2}$: 1.1×10^{-23} in a terrestrial laboratory ($g = 9.81$), 3.1×10^{-17} at Earth's orbital acceleration, 2.6×10^{-13} at Saturn's, and of order unity only in the outer disc of a galaxy. The g^{-2} scaling is exact to numerical precision.

This is a falsifiable statement, and a distinctive one: it is the **opposite** of a constant SME background. The violation is largest where no Lorentz test exists and smallest where the tests are tightest. A search that bins Lorentz-violation limits by the local gravitational acceleration would test it directly. It is not confronted with data here.

8. What is not derived

The kernel's first moment must satisfy $M_1 = (2/3)c/a_0$ once the memory force of (12) is included — numerically $2.13 \times 10^{18} \text{ s} \approx 68 \text{ Gyr}$. (Before that correction it was $c/a_0 = 3.20 \times 10^{18} \text{ s} \approx 101 \text{ Gyr}$, which on the pure- Λ footing is Z/H_Λ with $Z = 2\sqrt{8\pi/3}$, because $a_0 = cH_\Lambda/Z$ there.)

In the short-memory limit **only** M_1 survives: three different kernel shapes (exponential, gamma-2, box) normalised to the same first moment give the same Θ to eight digits. The shape is invisible; one number has been traded for one number. This is the same reparametrisation already known to afflict the temperature-functional class, in which the crossover is $a_0/(cH_\Lambda) = 2/r$ with r a free ratio of two slopes of the generating function.

Additionally the kernel memory time λ is a new constant of the construction, and it is bounded. Short memory must hold wherever the *acceleration* dependence does physical work — that is, where MOND is tested. Requiring $x = \lambda\Omega \leq 0.1$ across the galactic range gives only $\lambda \leq 0.98 \text{ Myr}$. In the solar system the theory need merely be Newtonian, which the long-memory branch can be: there $\Theta \rightarrow 4v/(\pi a_0 \lambda)$, and the residual anomaly is

$$(22) \quad \Delta = g(1 - \mu) \simeq g/(4\Theta^2) = g \pi^2 a_0^2 \lambda^2 / (64 v^2)$$

Imposing $\Delta \leq 3.66 \times 10^{-14}$ planet by planet, the tightest is **Mercury**, giving

$$(23) \quad \lambda \leq 1.24 \times 10^9 \text{ s} \approx \mathbf{39 \text{ years}}, \text{ hence } N = (c/a_0)/\lambda \geq 2.6 \times 10^9$$

At that λ the Milky Way sits at $x = 1.1 \times 10^{-6}$, deep in short memory, so $\Theta = |a|/a_0$ holds exactly where the radial acceleration relation is measured. A galaxy-scale λ remains excluded outright: at

$\lambda = 35$ Myr, Earth falls in the long-memory branch with $\Theta = 0.37$, a tens-of-percent inertia shift.

Therefore: the coefficient relating a_0 to Λ is fitted, not derived. On the pure- Λ footing the relation $a_0 = \frac{1}{2}c \sqrt{(G\rho_\Lambda)} = c^2 \sqrt{(\Lambda/32\pi)}$ is an input to this construction, and the rational $\frac{1}{2}$ has no derivation here. The author has previously and publicly retracted broader claims; none are made here.

9. Summary and open problems

Established: a parity theorem excluding all polynomial-in- u worldline self-interactions (Theorem 2 in the text numbering above is the rest-energy theorem; the parity result is Theorem 1); the rapidity gap as the unique escape class; a closed-form reduction of a causal one-scale kernel; the MI equation of motion; the $\alpha = 2$ interpolation in closed form; a structural exclusion of the speed-dependent branch; the two-function repair of the rest-energy defect; the general-orbit equation of motion with its memory force; and the $\alpha = 2$ cancellation of the Newtonian anomaly.

Open:

1. **A ghost analysis of the nonlocal theory.** The equation of motion (9) is fourth order in x . Ostrogradsky's theorem does not apply, because it requires a *local* higher-derivative Lagrangian and (7) is first-order-nonlocal; but that is an argument for why the theorem is silent, not a proof that the nonlocal theory is ghost-free. That proof is not attempted here and is the sharpest remaining structural gap.
2. **The SME phenomenology of (20).** The surviving structure is CPT-even of size $(1-\mu)/2 \approx 10^{-23}$ in a terrestrial laboratory, comfortably below matter-sector $c^{\{\mu\nu\}}$ bounds, but a full confrontation binned by local g (the g^{-2} signature of §7) has not been carried out.
3. **Quantisation.** The $|s|$ in (4) means the action admits no local derivative expansion, so there is no effective-field-theory power counting and no controlled quantum completion by that route. This is the flip side of the feature: $|a|$ is precisely the non-analytic object MOND requires.
4. **The coefficient.** $M_1 = (2/3)c/a_0$ remains an input.

Reproducibility

Every quantitative claim above is generated by a committed, runnable script that exits non-zero on any failed check and includes negative controls that must trip:

Script	Checks	Content
<code>mi_nonquadratic_u_escape_20230630.py</code>	30/30	parity theorem; the rapidity-gap identities (3)–(6)
<code>mi_rapidity_kernel_solved_20230630.py</code>	35/35	closed form (8); the EOM (9); F in closed form (10); λ bound
<code>mi_two_function_restmass_fix32026.py</code>	32/32	Theorem 2; Forms I and II; (12)–(16)
<code>mi_ephemeris_and_action_pinc382026.py</code>	38/38	the α -family anomaly Δ ; the $\alpha \geq 1.4$ bound

Script	Checks	Content
mi_noncircular_ctp_eom_2026.py	37/27	general orbits (10)–(13); the memory force (9); the $\alpha = 2$ cancellation (15)
mi_form3_cpt_even_and_lambda_bound_2026.py	28/28	Form III (15); the antimatter kill (14); the λ bound (17)–(18)
mi_ctp_variational_2026.py	50/50	the in-in variational property of the retarded class
mi_crossover_master_formula_2026.py	24/24	the $a_0/(cH\Lambda) = 2/r$ reparametrisation

10. Changes in versions 2 and 3

Version 3

1. **§5 corrected — this is the substantive change.** v1 and v2 both asserted that on a circular orbit $|a|$ is constant and so “the acceleration dependence contributes no additional radial force.” **That was wrong.** $\hat{a} = a/|a|$ rotates, $d^2\hat{a}/dt^2 = \Omega^2 r \neq 0$, and the memory force is outward. The correct balance is (12), $\bar{g} = g_{\text{obs}}[\mu + (Y/2)\mu]$. MOND survives as a renormalisation $a_0 \rightarrow (2/3)a_0$, so $M_1 = (2/3)c/a_0$, and any target interpolation is still exactly reachable via (13).
2. **General orbits solved** (10)–(11): $\theta(\tau, \tau-s) = (s/c)|a(\tau-s/2)|$, so Θ is a retarded memory-average of $|a|$ with lag $s/2$. The in-in prescription’s role is now quantified: the advanced terms enter at $O(\lambda\Omega) \leq 4 \times 10^{-6}$ galactically, so (9) is causal at the accuracy that matters.
3. **New result** (15): the memory force cancels the leading Newtonian anomaly identically at $\alpha = 2$, suppressing the solar-system residual by $8Y^2 = 3.2 \times 10^{16}$ to $1.2 \times 10^{-35} \text{ m s}^{-2}$. Against interest: the $\alpha = 1$ floor survives as $a_0/4$ (still excluded, $640\times$), α_{min} softens only to 1.322, and above $\alpha = 2$ the residual flips sign.
4. §9 item 1 replaced: non-circular orbits are no longer open; a ghost analysis of the nonlocal theory is, and is now the sharpest remaining structural gap.

Version 2

1. **§6 rewritten.** v1 presented two repairs and left the CPT-odd one owing an SME analysis. That framing was wrong. The correct covariant object is $\sqrt{((u \cdot n)^2)}$; with it the repair is CPT-even *and* bounded below (Form III, eq. 20), so Form I’s 0.8165 c instability was an unnecessary price and Form II’s CPT problem was an artefact of the writing. The owed SME analysis is discharged by the antimatter argument (eq. 14), which excludes the odd branch outright.
2. **§8 bound corrected, against the paper’s own conservatism.** v1 stated $\lambda \leq 1.4$ days and $N \geq 2.6 \times 10^{13}$, derived by demanding short memory at Mercury. That requirement was too strong: the solar system need only be Newtonian, not acceleration-resolved. The correct bound from the ephemeris anomaly is $\lambda \leq 39$ years and $N \geq 2.6 \times 10^9$ — weaker by a factor 1.0×10^4 .
3. Script table extended; §9 item 2 restated.

4. Unchanged: a_0 is not derived, there is no local derivative expansion, and the coefficient of a_0 remains fitted.

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